

Ê PHASOR TEMPLATES (Air: ε_r=μ_r=1)

Wavenumber per medium *nonmag.*

$$k_i = \frac{\omega\sqrt{\epsilon_{r,i}}}{c} \text{ (rad/m)} \Rightarrow k_2 = k_1\sqrt{\epsilon_{r,2}/\epsilon_{r,1}}$$

Direction table *u* ∈ {*x*, *y*, *z*} = *axis perp. to boundary*

Inc dir	Inc	Refl	Trans
+û	e^{-jk_1u}	e^{+jk_1u}	e^{-jk_2u}
-û	e^{+jk_1u}	e^{-jk_1u}	e^{+jk_2u}

+û direction ⇔ Med 2 at u ≥ 0. Reflected sign flips, transmitted matches incident.

General template (incident has pol. *ê*, amplitude *a*₀):

Êⁱ = *a*₀ê *e*^{∓*jk*₁*u*} (V/m) (incident wave, Med 1)

Ê^r = Γ *a*₀ê *e*^{±*jk*₁*u*} (V/m) (reflected wave, Med 1)

Ê^t = τ *a*₀ê *e*^{∓*jk*₂*u*} (V/m) (transmitted wave, Med 2)

(lossy med 2: replace *e*^{∓*jk*₂*u*} with *e*^{∓α₂*u*} *e*^{∓*jk*₂*u*})

Total in medium 1: Ê₁ = Êⁱ + Ê^r.

ê by polarization *plug into templates above*

Pol.	ê	Note
Lin. <i>x̂</i>	<i>x̂</i>	<i>E_y</i> = 0
Lin. <i>ŷ</i>	<i>ŷ</i>	<i>E_x</i> = 0
Lin. at 45°	(<i>x̂</i> + <i>ŷ</i>)/√2	in phase
RHC	<i>x̂</i> - <i>jŷ</i>	δ = -π/2
LHC	<i>x̂</i> + <i>jŷ</i>	δ = +π/2
General	<i>a_xx̂</i> + <i>a_yŷ</i> <i>e^{jδ}</i>	elliptical

Γ, τ act on ê unchanged. Only swap the polarization, not the formulas.

TABLE 8-2 — Γ, τ, R, T SUMMARY

Med. 1 (*z* < 0) incident side; Med. 2 (*z* ≥ 0) transmitted.

	Normal⊥ pol (θ _i =0)	∥ pol
Γ refl. wave	$\frac{\eta_2 - \eta_1}{\eta_2 + \eta_1}$	$\frac{\eta_2 \cos \theta_i - \eta_1 \cos \theta_t}{\eta_2 \cos \theta_i + \eta_1 \cos \theta_t}$
τ trans. wave	$\frac{2\eta_2}{\eta_2 + \eta_1}$	$\frac{2\eta_2 \cos \theta_i}{\eta_2 \cos \theta_i + \eta_1 \cos \theta_t}$
τ rel.	$\tau = 1 + \Gamma_{\perp}$	$\tau_{\parallel} = (1 + \Gamma_{\parallel}) \frac{\cos \theta_i}{\cos \theta_t}$
R refl. pwr	$ \Gamma ^2$	$ \Gamma_{\perp} ^2$
T trans. pwr	$ \tau ^2 \frac{\eta_1}{\eta_2}$	$ \tau_{\parallel} ^2 \frac{\eta_1 \cos \theta_t}{\eta_2 \cos \theta_i}$
R+T total	$T = 1 - R$	$T_{\perp} = 1 - R_{\perp}$

Notes: sin θ_t = √μ₁ε₁/μ₂ε₂ sin θ_i; nonmag. ⇒ η₂/η₁ = n₁/n₂.

Intrinsic impedance η_i *plug into table above*

$$\eta_i = \sqrt{\frac{\mu_r}{\epsilon_r}} \eta_0 \xrightarrow{\mu_r=1} \frac{\eta_0}{\sqrt{\epsilon_{r,i}}} \text{ (}\Omega\text{)} \quad \eta_0 = 120\pi \approx 377 \Omega$$

Default μ_r = 1 (air, dielectric). Use μ_r ≠ 1 only if problem says “ferromagnetic”/iron/ferrite/etc.

Low-loss correction η_c σ/(ωε) ≪ 1

$$\eta_c \approx \sqrt{\frac{\mu}{\epsilon'}} \left(1 + j \frac{\sigma}{2\omega\epsilon'}\right) \xrightarrow{\text{drop } j} \frac{\eta_0}{\sqrt{\epsilon_r}}$$

For Γ/τ just use η₀/√ε_r. The *j* term is for |η_c|, θ_η. See LOSSY MEDIA — 5 CASES for all regimes.

POWER REFLECTION & TRANSMISSION

% reflected *always*

$$\%P_r = 100 |\Gamma|^2$$

% transmitted *η-ratio matters!*

$$\%P_t = 100 |\tau|^2 \frac{\eta_1}{\eta_2}$$

Check: %P_r + %P_t = 100.

SNELL'S LAW — OBLIQUE INCIDENCE

Single boundary *angles from normal*

$$\sin \theta_2 = \sin \theta_1 \sqrt{\frac{\epsilon_{r,1}}{\epsilon_{r,2}}} = \sin \theta_1 \frac{n_1}{n_2}$$

Multilayer chain *stack of slabs 1 → 2 → 3 → 4*

$$\sin \theta_{i+1} = \sin \theta_i \sqrt{\frac{\epsilon_{r,i}}{\epsilon_{r,i+1}}}$$

Air–slabs–air shortcut *outer media identical*

$$\theta_{\text{exit}} = \theta_{\text{incident}}$$

Snell only sets angles. Reflection/transmission magnitudes need Fresnel coefficients (parallel vs perpendicular).

LATERAL DISPLACEMENT

Beam offset through *N* slabs thickness *t_i*, refracted angle inside slab *i*

Slab-indexed θ_{*i*} = angle inside slab *i*; *t_i* = slab thickness (*m*)

$$d = \sum_{i=1}^N t_i \tan \theta_i$$

The angle (θ_{*i*}) in the formula is always the one the ray makes inside that slab of height (*t_i*).

LOSSY MEDIA — 5 CASES

Loss tangent ε' = ε_rε₀; ε₀ = 8.854 × 10⁻¹² F/m

$$\frac{\epsilon''}{\epsilon'} = \frac{\sigma}{\omega \epsilon_r \epsilon_0}$$

Type	σ/ωε	η _c , α, β
Lossless (σ=0)	0	η = η ₀ /√ε _r exact α = 0, β = ω√ε _r /c
Low-loss	< 10 ⁻²	η _c ≈ $\frac{\eta_0}{\sqrt{\epsilon_r}} \left(1 + j \frac{\sigma}{2\omega\epsilon'}\right)$ α ≈ $\frac{\sigma\eta_0}{2\sqrt{\epsilon_r}}$, β ≈ $\frac{\omega\sqrt{\epsilon_r}}{c}$
Quasi-cond.	10 ⁻² –10 ²	exact: (η/√ <i>X</i>) <i>e^{jθ}</i> η see below
Good cond.	> 10 ²	η _c ≈ (1 + <i>j</i>) √π <i>f</i> μ / σ α ≈ β ≈ √π <i>f</i> μ σ
Magnetic	any	keep full √μ _r /ε _r η ₀

For Γ/τ: drop *j* correction; use η₀/√ε_r. Magnetic: keep μ_r.

Low-loss quick params σ/(ωε) ≪ 1, μ_r = 1

$$\alpha \approx \frac{\sigma \eta_0}{2\sqrt{\epsilon_r}} \text{ (Np/m)}, \quad \beta \approx \frac{\omega\sqrt{\epsilon_r}}{c} \text{ (rad/m)}, \quad \eta_c \approx \frac{\eta_0}{\sqrt{\epsilon_r}} \text{ (}\Omega\text{)}$$

Exact (quasi-cond) *X* = √[1 + (ε''/ε')²]

$$\alpha = \omega \sqrt{\frac{\mu \epsilon'}{2}} \sqrt{X-1}, \quad \beta = \omega \sqrt{\frac{\mu \epsilon'}{2}} \sqrt{X+1}$$
$$\theta_{\eta} = \frac{1}{2} \arctan \frac{\sigma}{\omega \epsilon'}$$

RECTANGULAR WAVEGUIDE — MODES

TE mode *E_z* = 0, *H_z* ≠ 0

TM mode *H_z* = 0, *E_z* ≠ 0

Dimensions *a* × *b* with *a* > *b* along *x̂*, *ŷ*; wave in *ẑ*.

E_x form (TE) *read m, n from arguments*

$$E_x \propto \cos\left(\frac{m\pi x}{a}\right) \sin\left(\frac{n\pi y}{b}\right) \sin(\omega t - \beta z)$$

coef. on *x* = *m*π/*a*; coef. on *y* = *n*π/*b*.

Identify TE_{*m**n*} vs TM_{*m**n*} *cos/sin pattern of E_x is identical for both!*

1. Read the problem statement (e.g. “a TE wave” → TE).

2. If *m* = 0 or *n* = 0 in the mode label ⇒ must be TE (TM forbids 0).

3. Source field *H_z*(*x*, *y*) given ⇒ TE. Source field *E_z*(*x*, *y*) given ⇒ TM.

TE allows at least one of *m*, *n* ≥ 1 (other can be 0). TM needs both *m*, *n* ≥ 1. That's why TE₁₀ exists but TM₁₀ doesn't.

TABLE 8-3 — FULL FIELD COMPONENTS

All fields carry an *e^{-jβz}* factor (omitted below). *k_c²* = (mπ/*a*)² + (nπ/*b*)².

TE mode *generator: Ê_z*

$$\hat{H}_z = H_0 \cos \frac{m\pi x}{a} \cos \frac{n\pi y}{b}$$
$$\hat{E}_x = -\frac{j\omega\mu}{k_c^2} \left(\frac{n\pi}{b}\right) H_0 \cos \frac{m\pi x}{a} \sin \frac{n\pi y}{b}$$
$$\hat{E}_y = -\frac{j\omega\mu}{k_c^2} \left(\frac{m\pi}{a}\right) H_0 \sin \frac{m\pi x}{a} \cos \frac{n\pi y}{b}$$
$$\hat{E}_z = 0, \quad \hat{H}_x = -\hat{E}_y/Z_{TE}, \quad \hat{H}_y = \hat{E}_x/Z_{TE}$$

TM mode *generator: Ê_z*

$$\hat{E}_z = E_0 \sin \frac{m\pi x}{a} \sin \frac{n\pi y}{b}$$
$$\hat{E}_x = -\frac{j\beta}{k_c^2} \left(\frac{m\pi}{a}\right) E_0 \cos \frac{m\pi x}{a} \sin \frac{n\pi y}{b}$$
$$\hat{E}_y = -\frac{j\beta}{k_c^2} \left(\frac{n\pi}{b}\right) E_0 \sin \frac{m\pi x}{a} \cos \frac{n\pi y}{b}$$
$$\hat{H}_z = 0, \quad \hat{H}_x = -\hat{E}_y/Z_{TM}, \quad \hat{H}_y = \hat{E}_x/Z_{TM}$$

TEM (plane wave) for reference: Ê_{*x*} = *E*₀*e^{-jβz}*, Ê_{*y*} = Ê_{*x*}/η, *f_c* = *N*/*A*, *k* = ω√μ̵.

CUTOFF FREQUENCY

f_{mn} (a.k.a. *f_c* for that mode) *m, n integers ≥ 0 (not both zero)*

$$f_c \equiv f_{mn} = \frac{u_{p0}}{2} \sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2} \text{ (Hz)}$$

Common cutoffs (*a* > *b*)

Mode	<i>f_{mn}</i>
<i>u_{p0}</i>	<i>c</i> /√ε _r (hollow: <i>u_{p0}</i> = <i>c</i>)
TE ₁₀	<i>f</i> ₁₀ = <i>u_{p0}</i> /(2 <i>a</i>) (dominant)
TE ₀₁	<i>f</i> ₀₁ = <i>u_{p0}</i> /(2 <i>b</i>)
TE ₂₀	<i>f</i> ₂₀ = <i>u_{p0}</i> / <i>a</i> = 2 <i>f</i> ₁₀
TE ₁₁ , TM ₁₁	<i>f</i> ₁₁ = $\frac{u_{p0}}{2} \sqrt{1/a^2 + 1/b^2}$

Propagation requires *f* > *f_{mn}*. In *Z_{TE}*/*Z_{TM}* and *u_g*, *f_c* = *f_{mn}* of the excited mode.

ε_r, FROM DISPERSION

Given ω, β, mode (*m*, *n*) *result is unitless*

$$\epsilon_r = \frac{c^2}{\omega^2} \left[\beta^2 + \left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2 \right] \text{ (unitless)}$$

Inputs: ω (rad/s), β (rad/m), *a*, *b* (m), *c* = 3 × 10⁸ (m/s), *m*, *n* (unitless).

WAVE IMPEDANCE & H_y

$$Z_{TE} = \frac{\eta}{\sqrt{1 - (f_c/f)^2}} \text{ (}\Omega\text{)} \quad Z_{TM} = \eta \sqrt{1 - (f_c/f)^2}$$
$$\eta = \frac{\eta_0}{\sqrt{\epsilon_r}} \quad H_y = \frac{E_x}{Z_{TE}} \text{ (TE mode)}$$

Z_{TE} > η always (denominator < 1); *Z_{TM}* < η.

GROUP VELOCITY & TRAVEL TIME

$$u_g = u_{p0} \sqrt{1 - (f_{mn}/f)^2} \text{ (m/s)}$$
$$u_p = \frac{u_{p0}}{\sqrt{1 - (f_{mn}/f)^2}} \text{ (m/s)} \Rightarrow u_p u_g = u_{p0}^2 \text{ (m}^2\text{/s}^2\text{)}$$

t = *L*/*u_g* (s) travel time through guide of length *L*

Higher modes have lower *u_g* ⇒ they arrive later. Group delay spread = sum over excited modes.

BOUNDARY

η₁, η₂ — imp. (Ω)

Γ, τ, *R*, *T* — coeffs (unitless)

τ = 1 + Γ; *R* = |Γ|²

k_i = ω√ε_{*r,i*}/c (rad/m)

α₂ (Np/m), β₂ (rad/m): lossy

z = 0 — boundary plane (m)

η₀ = 120π ≈ 377 Ω

SNELL / OBLIQUE

θ_{*i*}, θ_{*t*} — angles from normal

n_i = √ε_{*r,i*} — index of refr.

t_i — slab thickness (m)

d — lateral disp. (m)

plane of inc. = plane of *k̂* + normal

⊥: Ê ⊥ this plane

∥: Ê ∥ this plane

WAVEGUIDE

a, *b* — wide, narrow dim. (m); *a* > *b*

m, *n* — mode indices (int.)

f_{mn} — cutoff (Hz)

β — prop. const (rad/m)

Z_{TE}/*Z_{TM}* — wave imp. (Ω)

E_x, *H_y* — field comps (V/m, A/m)

η = η₀/√ε_r

VELOCITIES

u_{p0} = *c*/√ε_r — bulk phase vel.

u_p = *u_{p0}*/√[1 - (*f_c*/*f*)²]

u_g = *u_{p0}*√[1 - (*f_c*/*f*)²]

u_p *u_g* = *u_{p0}*²

t = *L*/*u_g* — travel time

c = 3 × 10⁸ m/s

POWER & UNITS

%P_r = 100|Γ|²

%P_t = 100|τ|² η₁/η₂

%P_r + %P_t = 100

σ (S/m), ε (F/m), μ (H/m)

ε₀ = 8.85 × 10⁻¹² F/m

μ₀ = 4π × 10⁻⁷ H/m

1/(36π) × 10⁻⁹ ≈ ε₀